Quick Start

From a cold start to a costed passage

Transit Calculator

Version 1.0 · 14 August 2026

> **Who this is for**
> Anyone who wants a transit computed today. No prior knowledge of the tool is assumed. Ten minutes end to end.

1 Start it

Double-click the "Transit Calculator" shortcut on the desktop. If there is no shortcut, run start_transit.bat in the project folder, or:

```
python server.py
```

From the project folder. Add --port 8079 to run a second copy alongside the first.

A browser opens on the console once the server is actually listening. It binds to your own machine only — nothing is exposed to the network. If the port is already in use it says so and stops, rather than starting a second server on top of the first.

2 Get a line on the chart

You need a track before anything can be computed. Either:

- DRAW ONE — click the chart to drop points. Drag a point to move it; click one and press Delete to remove it; Undo steps back.
- LOAD ONE — pick a file under "Load a line" and press Load. Shapefile, GeoJSON, GPX and CSV all work. A shapefile needs its .prj, or drop the whole set in as a .zip.

The supplied survey_transit_line is already in the list. Load it and the status bar reads 77.68 NM over 6 legs.

> **Check which way it runs**
> The ends are labelled START and END, and chevrons along the track show the direction of travel. Press REVERSE — at the foot of the left panel, just above the offline downloads — to run it the other way. This is not cosmetic: with a wind on the nose one way and astern the other, the two directions can differ by a fifth of the fuel.

3 Set the passage

Field	What it means	Start with
Speed kt	How fast the vessel goes	8
Speed is	Through water = a water speed to hold, and the current then decides your ETA. Over ground = a fixed arrival, and the current decides the fuel. Fixed revs = you set the throttle and the speed is the answer.	through water
Revs	Fixed-revs mode only, replacing the speed box: the RPM the vessel is holding.	—
Depart UTC	When you leave. The tide is different three hours later, so this genuinely changes the answer.	as offered
Configuration	Which vessel configuration is fitted	as offered
Fuel aboard	Litres in the tank at departure. Leave it blank and a full tank is assumed.	blank
Currents	Leave ticked. "all models (chained)" is the right default.	all models
Weather	IDW over the line pulls real buoy and forecast data.	IDW over the line

Every control carries a tooltip, and so does every abbreviation the console shows you — the column headings, the result chips, the model names, the cycle tags. Hover one for what it means and why it matters, without leaving the console for this manual.

4 Calculate

Press CALCULATE. First run takes a few seconds while the weather is fetched; afterwards it is quick. You get four headline numbers — distance, time, litres, average speed over ground — then a row per leg.

5 Read the flags before you read the numbers

The chips under the headline are the honest part of the tool. They tell you how much of the answer rests on real data.

Chip	Meaning	What to do
current 100% covered	Every step of the track had a real forecast current.	Nothing. This is what you want.
current 87% covered · 10.4 NM gap	Part of the line is outside every cached model. Those miles were computed with ZERO current and counted.	Press Currents under "Localize for offline" — it works out which models the line needs and fetches them.
20% projected	A cached cycle did not reach your departure time, so the current was projected forward by whole tidal cycles.	Re-fetch Currents for a fresh cycle.
RPM outside fitted window	The speed you asked for needs revs outside the range the fuel law was fitted over.	The fuel figure is an extrapolation. Treat it as indicative.
DBOFS 87% + CBOFS 13%	Two current models were chained to cover the line.	Nothing — this is the system working.
140 L aboard	You stated a part-full tank, so the margin is against that and not against a full one.	Nothing, if the figure is right. It is amber because it is an assumption you typed, not one the tool can check.
FUEL SHORT BY 12 L	The passage burns past the reserve floor.	Load more fuel, slow down, or shorten the line.

6 Export it

Name it, choose the shapefile CRS, and press a format:

- Shapefile (.zip) — polyline + .prj/.dbf/.shx/.cpg
- Shapefile — per leg (.zip) — one feature per leg, with results
- GeoJSON — WGS84 lon/lat, track + waypoints
- KML — Google Earth
- GPX — route + track, for a chartplotter
- CSV — waypoints — lat/lon, leg and cumulative NM
- CSV — leg table — the computed transit

The two "per leg" exports carry the computed results as attributes, so the GIS user gets speed, set, drift, fuel and ETA attached to the geometry rather than in a separate table to join by hand.

Where next

- Instruction Manual — every control and field in detail.
- Technical Manual — how the numbers are produced, and their limits.
- Development Manual — the code, the tests, and how to extend it.